Topology, Limits & Continuity — 80/20 Interview Notes

0. The Big Picture

The most important mental model is:

Real Numbers

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▼

Sequences

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▼

Limits

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▼

Continuity

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▼

Differentiability

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▼

Gradients

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▼

Gradient Descent

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▼

Machine Learning

The chapter explicitly connects sequences → limits → gradients → gradient descent → model training.

Interview takeaway

If an interviewer asks:

Why do we care about topology, limits and continuity in machine learning?

A strong answer is:

They provide the mathematical foundation for optimization. Limits allow us to define derivatives, continuity gives functions predictable local behavior, and differentiability allows gradients to be computed. Gradient-based optimization such as gradient descent then uses those gradients to adjust model parameters.

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1. Topology

Definition

Topology studies the **local properties of spaces and how their parts are organized or connected**.

For this chapter, the most important concepts are:

- open sets
- closed sets
- neighborhoods
- sequences
- boundedness
- infimum/supremum
- compactness
- limits
- continuity

80/20 intuition

Don't think of topology as something you immediately need to calculate.

Think:

Topology gives us the language for talking about "nearby", "inside", "boundary", and "convergence".

This becomes important when we ask:

What happens to a function when its input gets arbitrarily close to some point?

That question is the beginning of limits.

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2. Open Sets

Definition

A set $A \subseteq \mathbb{R}$is **open** if for every point $x \in A$, we can find some $\epsilon > 0$such that

$$(x, -\epsilon x + \epsilon) \subseteq A$$

In simpler language:

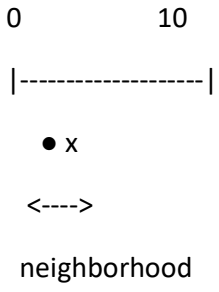
Every point inside an open set has some small neighborhood that remains completely inside the set.

This is the formal definition given in the chapter.

Diagram

Consider:

A = (0, 10)

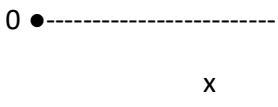


The neighborhood around x

can stay completely inside A.

But consider the endpoint:

A = (0, 10)



Any neighborhood around x

extends outside A.

Therefore x cannot have a

neighborhood completely inside A.

That's why:

$$(0, 10)$$

is open.

Example

$$A = (2, 5)$$

Take:

$$x = 3$$

We can choose:

$$\epsilon = 0.5$$

Then:

$$(2.5, 3.5) \subset (2, 5)$$

Therefore 3 has a neighborhood completely inside the set.

---

3. Closed Sets

Definition

A set is **closed** if its complement is open.

For example:

$$[0, 1]$$

is closed because

$$\mathbb{R} \setminus [0,1] = (-,\infty 0) \cup (1,\infty)$$

which is open.

Important interview distinction

Do NOT think:

"If something isn't open, it must be closed."

False.

For example:

$$(0,1]$$

is neither open nor closed.

Quick table

Set	Open?	Closed?
$(0,1)$	Yes	No
$[0,1]$	No	Yes
$(0,1]$	No	No
$[0,1)$	No	No
$\mathbb{R}$	Yes	Yes
$\emptyset$	Yes	Yes

Interview question

Q: Can a set be both open and closed?

Yes.

For example:

$$\mathbb{R}$$

is both open and closed under these definitions.

---

4. Neighborhood

A neighborhood of $x$is essentially an open interval surrounding x.

x

●

<------->

ε

(x-ε, x+ε)

80/20 meaning

A neighborhood means:

"Let's look sufficiently close to this point."

This idea appears repeatedly in:

- limits
- continuity
- derivatives
- optimization

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5. Open Ball

The one-dimensional neighborhood can be generalized using distance.

The chapter defines:

$$B(x,r) = \{y \in V : \| x - y \| < r\}$$

as a ball of radius $r$centered at x.

For one dimension:

$$B(x,r) = (x, -rx + r)$$

because:

$$\| x - y \| = | x - y |$$

Diagram

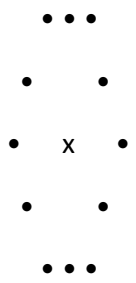
1D

x-r x x+r

|----------●----------|

radius = r

2D



This becomes important when moving from:

$$\mathbb{R}$$

to:

$$\mathbb{R}^n$$

which is where machine-learning parameters normally live.

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6. Closed Sets and Sequences

This is a particularly important interview concept.

A set $A$is closed iff:

Whenever a sequence stays inside $A$and converges, its limit is also inside A.

The chapter states this as the characterization of closed sets using sequences.

Example

Consider:

$$A = [0, 1]$$

Take:

$$a_n = \frac{1}{n}$$

Every $a_n$belongs to A.

And:

$$a_n \to 0$$

Since:

$$0 \in [0, 1]$$

the limit remains inside the set.

Contrast

Consider:

$$A = (0, 1)$$

Again:

$$a_n = \frac{1}{n}$$

Every:

$$a_n \in A$$

but:

$$\lim_{n \to \infty} a_n = 0$$

and:

$$0 \notin (0, 1)$$

Therefore $(0, 1)$is not closed.

Interview answer

A closed set contains the limits of all convergent sequences whose elements belong to that set.

This is much more useful to remember than memorizing only "complement is open."

---

7. Bounded Sets

Definition

A set $A \subseteq \mathbb{R}$is:

Bounded below

There exists $m$such that:

$$m \leq x$$

for every $x \in A$.

Bounded above

There exists Msuch that:

$$x \leq M$$

for every $x \in A$.

Bounded

Both conditions hold.

Diagram

For:

$$A = [2, 5]$$

----|---------|---------|----

2 A 5

↑ ↑

lower bound upper bound

More generally:

m ========== M

A

---

8. Minimum vs Infimum

This is a very common mathematical interview distinction.

Minimum

A minimum must actually belong to the set.

Example:

$$A = [0, 1]$$

Then:

$$\min A = 0$$

because:

$$0 \in A$$

Infimum

The infimum is the **greatest lower bound**.

Example:

$$A = (0, 1)$$

There is no minimum because 0 is not in the set.

But:

$$\inf A = 0$$

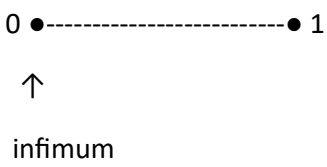
The chapter specifically highlights the example:

$$A = \{1/n : n \in \mathbb{N}\}$$

where 0 is the greatest lower bound but isn't an element of the set.

Diagram

A = (0,1)



0 is NOT part of A.

Therefore:

inf A = 0

but

min A does not exist.

---

9. Maximum vs Supremum

Same idea from the other side.

For:

$$A = (0, 1)$$
$$\sup A = 1$$

but:

$$\max A$$

doesn't exist.

Remember

MINIMUM → actually inside set

INFIMUM → greatest lower bound

MAXIMUM → actually inside set

SUPREMUM → smallest upper bound

Interview trap

Q: Is infimum always an element of the set?

No.

Q: Is minimum always an element of the set?

Yes.

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10. Why Bounds Matter in Machine Learning

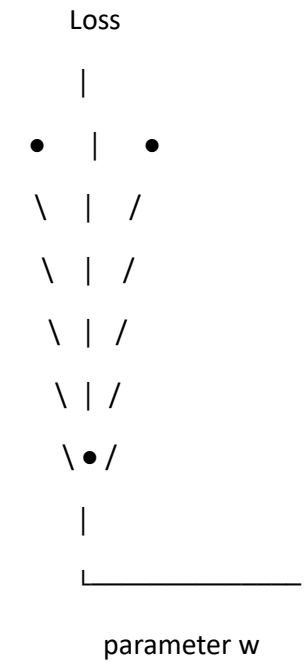
Machine learning can be viewed as an optimization problem.

The chapter describes the objective roughly as finding:

$$\inf_{w} Loss(f, (x, w)y)$$

over the parameter space.

Think:



We want to find a parameter configuration giving very small loss.

But there is an important distinction:

The infimum of a function does not necessarily have to be attained.

That is why compactness becomes useful.

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11. Compact Sets

Definition

In $\mathbb{R}$, the chapter defines a compact set as a set that is:

closed + bounded

Example:

$$[0, 1]$$

is compact.

But:

$$(0, 1)$$

is not compact because it is not closed.

And:

$$[0, \infty)$$

is not compact because it is unbounded.

80/20 intuition

Compactness roughly gives us:

A finite, contained, well-behaved search region.

For continuous functions, compactness gives a very important guarantee:

The function actually achieves its minimum and maximum.

The chapter states that a continuous function on a compact set is bounded and attains its extrema.

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12. Compactness → Optimization

This is worth remembering as a chain:

Closed + Bounded

|

▼

Compact set

|

▼

Continuous function

|

▼

Minimum / Maximum

actually exists

Interview question

Why is compactness useful in optimization?

Answer:

Compactness prevents the optimization domain from escaping to infinity or losing boundary points. For a continuous function defined on a compact set, the extreme value theorem guarantees that minimum and maximum values are actually attained.

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13. Bolzano-Weierstrass Theorem

The chapter states:

Every sequence in a compact set has a convergent subsequence.

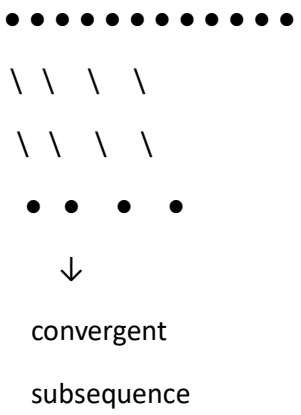
Formally:

$$A \text{ compact} \Rightarrow \text{every sequence in } A \text{ has a convergent subsequence.}$$

Diagram

Suppose:

Sequence:



The original sequence may bounce around, but we can extract a subsequence that converges.

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14. Limits

Now we reach one of the most important concepts.

Intuition

When we write:

$$\lim_{x \to a} f(x) = L$$

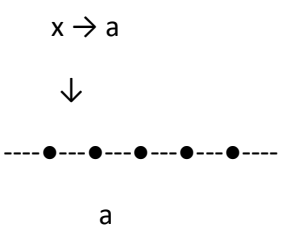
we mean:

As $x$gets arbitrarily close to a, $f(x)$gets arbitrarily close to L.

The chapter defines function limits through convergent sequences.

Diagram

x-values:



|

▼

f(x) values:

----●---●---●---●---●----

↓

L

The key idea is:

input approaches a

↓

output approaches L

---

15. Critical Interview Point: Limit ≠ Function Value

Consider:

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

Then:

$$\lim_{x \to 0} f(x) = 0$$

but:

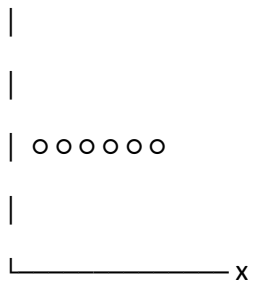
$$f(0) = 1$$

The chapter explicitly uses this example to show that a function's limit doesn't have to equal its value at the point.

Diagram

y

| ● (0,1)



The open circle represents the nearby behavior.

The isolated point represents $f(0) = 1$.

Interview question

Does the existence of a limit imply continuity?

No.

We additionally need:

$$\lim_{x \to a} f(x) = f(a)$$

---

16. Epsilon-Delta Definition

This is the formal definition of a limit.

$$\lim_{x \to a} f(x) = L$$

means:

For every:

$$\epsilon > 0$$

there exists:

$$\delta > 0$$

such that if:

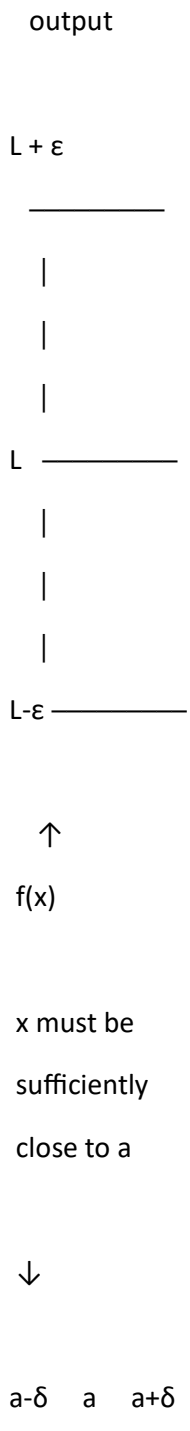
$$0 < | x - a | < \delta$$

then:

$$| f(x) - L | < \epsilon$$

The chapter presents this as the equivalent epsilon-delta formulation.

The most important mental picture



———●———

Meaning

You choose how accurate the output must be:

$$\epsilon$$

I respond by finding how close the input must be:

$$\delta$$

This is the essence of the definition.

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17. A Simple Epsilon-Delta Example

Prove:

$$\lim_{x \to 2} x = 2$$

We need:

$$| x - 2 | < \delta$$

and want:

$$| f(x) - 2 | < \epsilon$$

Since:

$$f(x) = x$$

we have:

$$| f(x) - 2 | = | x - 2 |$$

Therefore simply choose:

$$\delta = \epsilon$$

Then:

$$| x - 2 | < \delta \Rightarrow | f(x) - 2 | < \epsilon$$

Done.

---

18. Limit Operations

If limits exist, many operations can be performed directly.

For example:

$$\lim(f + g) = \lim f + \lim g$$
$$\lim(cf) = c \lim f$$
$$\lim(fg) = (\lim f)(\lim g)$$

and, when the denominator's limit is nonzero:

$$\lim \frac{f}{g} = \frac{\lim f}{\lim g}$$

These properties are explicitly listed in the chapter.

Interview shortcut

If:

$$\lim_{x \to a} f(x) = 2$$

and:

$$\lim_{x \to a} g(x) = 3$$

then:

$$\lim(f + g) = 5$$
$$\lim(fg) = 6$$
$$\lim(2f) = 4$$

and:

$$\lim(f/g) = 2/3$$

---

19. Big-O and Little-o

The chapter also introduces asymptotic notation for functions.

Big-O

$$g(x) = O\big(f(x)\big)$$

roughly means:

$g$does not grow faster than a constant multiple of f, near the relevant point or as $x \to \infty$.

Example:

$$3x^2 + 10x + 5 = O(x^2)$$

because the dominant growth is x^2.

Little-o

$$g(x) = o\big(f(x)\big)$$

means:

$g$becomes negligible compared with f.

For example:

$$x = o(x^2)$$

as:

$$x \to \infty$$

because:

$$\frac{x}{x^2} = \frac{1}{x} \to 0$$

Interview distinction

O(f)

↓

g is at most comparable to f

o(f)

↓

g becomes negligible relative to f

---

20. Continuity

Definition

A function f is continuous at a if:

$$\boxed{\lim_{x \to a} f(x) = f(a)}$$

This is the central definition in the chapter.

Intuition

Think:

Small changes in input produce small changes in output.

Input

x ——————————► f(x)

| |

| small change | small change

▼ ▼

x + small f(x) + small

---

21. Continuity vs Limit

This is an extremely common interview question.

Limit

Asks:

What value does $f(x)$ approach as x approaches a?

Continuity

Asks:

Does that approaching value equal the actual value $f(a)$?

So:

Limit:

x → a

↓

f(x) → L

Continuity:

x → a

↓

f(x) → L

↓

L = f(a)

Therefore:

Continuity = Limit exists and equals function value

---

22. Continuity Example

Consider:

$$f(x) = x^2$$

At $x = 2$:

$$\lim_{x \to 2} x^2 = 4$$

and:

$$f(2) = 4$$

Therefore:

$$\lim_{x \to 2} f(x) = f(2)$$

so f is continuous at 2.

---

23. Discontinuous Example

Consider:

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

We already know:

$$\lim_{x \to 0} f(x) = 0$$

but:

$$f(0) = 1$$

Therefore:

$$0 \neq 1$$

so the function is not continuous at 0.

---

24. Sequence Characterization of Continuity

An important equivalent definition is:

If:

$$a_n \to a$$

then continuity requires:

$$f(a_n) \to f(a)$$

The chapter explicitly states this equivalent sequence formulation.

Interview answer

A function is continuous at $a$iff for every sequence $a_n$converging to a, the sequence $f(a_n)$converges to $f(a)$.

---

25. Operations Preserving Continuity

If $f$and $g$are continuous at a:

Addition

$$f + g$$

is continuous.

Multiplication

$$fg$$

is continuous.

Division

$$\frac{f}{g}$$

is continuous if:

$$g(a) \neq 0$$

Composition

If $g$is continuous at a, and $f$is continuous at $g(a)$:

$$f \circ g$$

is continuous at a.

These properties are summarized in the chapter.

---

26. Why Neural Networks Care About Continuity

This is the ML connection you should be able to explain in an interview.

Suppose:

$$y = f(x, w)$$

where $w$represents model parameters.

Training attempts to find:

$$w^* = \arg\min_w Loss(w)$$

Gradient-based optimization relies on derivatives.

Derivatives are defined using limits:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Therefore:

Continuity

↓

well-behaved function behavior

Differentiability

↓

derivatives

Derivatives

↓

gradients

Gradients

↓

gradient descent

Gradient descent

↓

model training

The chapter explicitly describes continuity and differentiability as important properties for gradient-based neural-network optimization.

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27. Derivative Connection

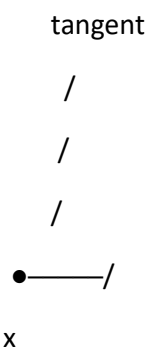
Although derivatives are formally covered in the next chapter, this chapter gives the definition:

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Or equivalently:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Diagram



The derivative tells us the **local slope**.

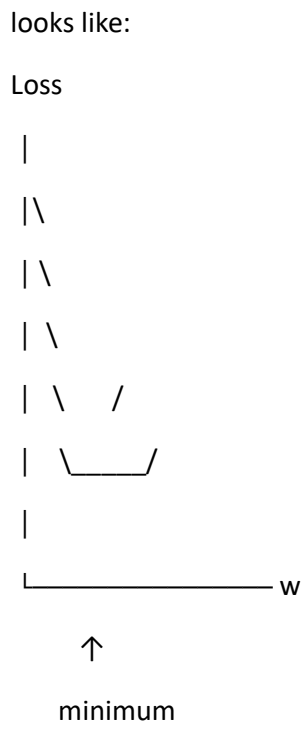
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28. Gradient Descent Connection

Suppose:

$$Loss(w)$$

looks like:



Gradient descent updates:

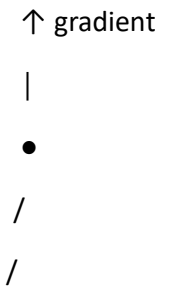
$$w_{new} = w_{old} - \eta \nabla Loss(w)$$

where:

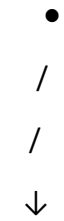
- w= model parameters
- $\nabla Loss(w)$= gradient
- η= learning rate

Intuition

If the gradient points uphill:



we move in the opposite direction:



---

29. Python Implementation — Numerical Limit

A practical implementation helps make the concept concrete.

```
def f(x):
    return x * x


a = 2


for h in [1, 0.1, 0.01, 0.001, 0.0001]:
    x = a + h
    print(h, f(x))
```

As $h \to 0$:

$$x \to 2$$

and:

$$f(x) \to 4$$

So:

$$\lim_{x \to 2} x^2 = 4$$

---

30. Python Implementation — Numerical Derivative

We can approximate:

$$f'(x)$$

using:

$$\frac{f(x+h)-f(x)}{h}$$

```
def f(x):
    return x ** 2


def derivative(f, x, h=1e-5):
    return (f(x + h) - f(x)) / h


print(derivative(f, 3))
```

Expected result is approximately:

6

because:

$$\frac{d}{dx}x^2 = 2x$$

and:

$$2(3) = 6$$

Interview connection

This is essentially showing how the derivative emerges from the limit definition.

---

31. Python Implementation — Gradient Descent

Consider:

$$f(w) = w^2$$

The derivative is:

$$f'(w) = 2w$$

Gradient descent:

$$w_{new} = w - \eta(2w)$$

Implementation:

```
w = 10.0
learning_rate = 0.1

for step in range(20):
    gradient = 2 * w
    w = w - learning_rate * gradient
    print(step, w)
```

The value of $w$moves toward:

$$0$$

which minimizes:

$$w^2$$

Conceptual flow

w = 10

|

▼

calculate loss

|

▼

calculate gradient

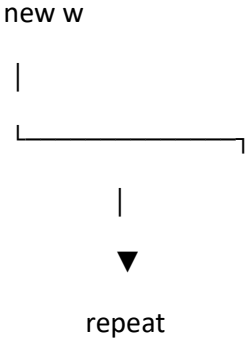
|

▼

move opposite gradient

|

▼



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32. A Very Important ML Example

Suppose a model has:

$$Loss(w) = (w - 5)^2$$

The minimum occurs at:

$$w = 5$$

Derivative:

$$Loss'(w) = 2(w - 5)$$

Starting with:

$$w = 0$$

gradient:

$$2(0 - 5) = -10$$

With learning rate:

$$\eta = 0.1$$

we get:

$$w_{new} = 0 - 0.1(-10) = 1$$

Next:

$$w = 1$$

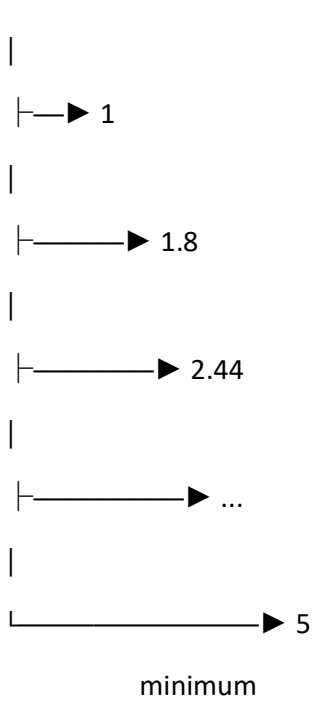
gradient:

$$2(1 - 5) = -8$$

Then:

$$w_{new} = 1.8$$

and so on.



This is the mathematical foundation behind optimization.

---

33. Important Special Limits

The chapter gives several useful limits.

1.

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1}$$

2.

$$\boxed{\lim_{x\to 0} x \log x = 0}$$

3.

For positive a:

$$\boxed{\lim_{x\to\infty} \frac{x^k}{e^{ax}} = 0}$$

for nonnegative integer k.

Intuition

Exponential growth eventually dominates polynomial growth.

x^2 → grows

x^3 → grows

x^{10} → grows

e^x → grows much faster

Therefore:

$$\frac{x^k}{e^{ax}} \to 0$$

---

34. Continuity of Elementary Functions

Elementary functions are continuous wherever they are defined.

Examples include:

- constants
- polynomial functions
- roots
- exponentials
- logarithms
- trigonometric functions

- inverse trigonometric functions
- hyperbolic functions

The chapter makes this observation explicitly.

Example

$$f(x) = \sin(x^2 + e^x) - 1 - 3x + 5x^4$$

is continuous wherever all operations are defined.

Interview shortcut

Instead of proving continuity from epsilon-delta every time:

Recognize that standard elementary functions are continuous on their domains, and sums/products/compositions of continuous functions remain continuous.

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35. The Most Important Theorem for Optimization

The chapter gives the following result:

A continuous function on a compact set attains its minimum and maximum.

In mathematical form:

If:

$$K$$

is compact and:

$$f: K \to \mathbb{R}$$

is continuous, then there exist:

$$\alpha, \beta \in K$$

such that:

$$f(\alpha) \leq f(x) \leq f(\beta)$$

for every:

$$x \in K$$

---

36. Why This Theorem Matters

Imagine:

$$f(x) = \frac{1}{x}$$

on:

$$(0, 1]$$

The function is continuous there, but as:

$$x \to 0^+$$

we get:

$$f(x) \to \infty$$

so it has no upper bound.

The chapter uses this as an example of what can go wrong without a closed-and-bounded domain.

Compare:

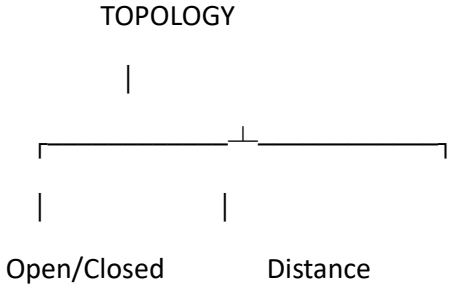
$$[0.1, 1]$$

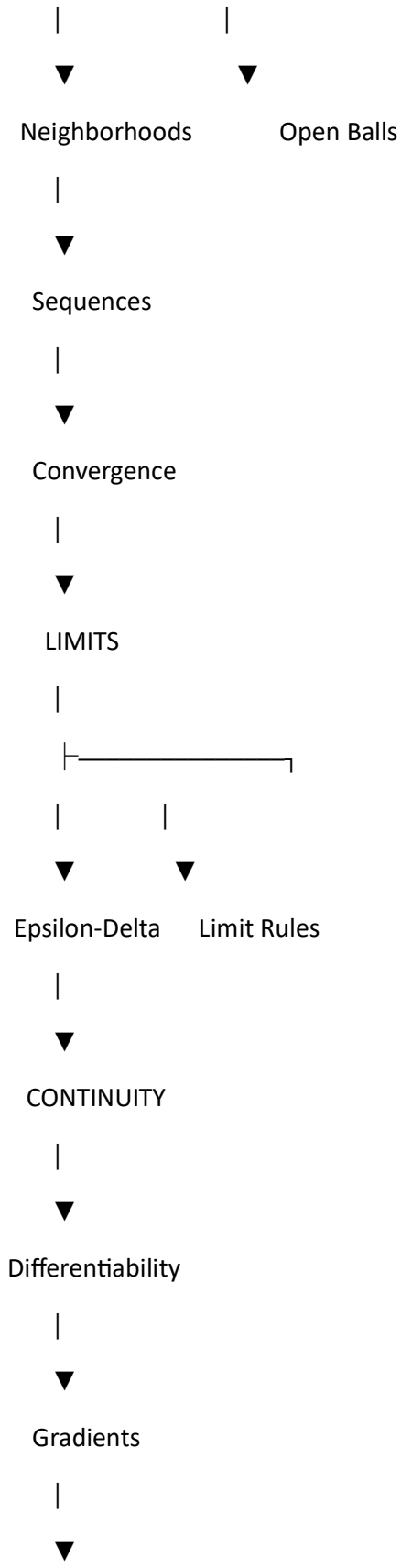
Now the problematic boundary at zero is excluded, and the domain is compact.

---

37. 80/20 Concept Map

For interview preparation, prioritize these:




Neighborhoods
Open Balls
Sequences
Convergence
LIMITS
Epsilon-Delta
Limit Rules
CONTINUITY
Differentiability
Gradients

Gradient Descent

|

▼

Machine Learning

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38. Interview Questions You Should Know

Q1. What is an open set?

Answer:

A set is open if every point in the set has a sufficiently small neighborhood that remains completely inside the set.

Q2. What is a closed set?

Answer:

A set is closed if its complement is open. Equivalently in $\mathbb{R}$, it contains the limits of all convergent sequences whose elements are in the set.

Q3. Can a set be neither open nor closed?

Yes.

Example:

$$(0, 1]$$

Q4. What is the difference between minimum and infimum?

Minimum must belong to the set.

Infimum is the greatest lower bound and doesn't necessarily belong to the set.

Example:

$$A = (0, 1)$$

has:

$$\inf A = 0$$

but no minimum.

Q5. What is the difference between maximum and supremum?

Maximum must belong to the set.

Supremum is the smallest upper bound and doesn't necessarily belong to the set.

Q6. What is a compact set?

For subsets of $\mathbb{R}$, a compact set is closed and bounded.

Q7. Why is compactness important?

Because continuous functions on compact sets attain their minimum and maximum.

Q8. What is a limit?

$$\lim_{x \to a} f(x) = L$$

means that $f(x)$can be made arbitrarily close to $L$by making $x$sufficiently close to a.

Q9. Does $f(a)$have to equal the limit?

No.

Example:

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

has:

$$\lim_{x \to 0} f(x) = 0$$

but:

$$f(0) = 1$$

Q10. What is continuity?

$$\boxed{f \text{ continuous at } a \iff \lim_{x \to a} f(x) = f(a)}$$

Q11. What is the epsilon-delta definition?

For every $\epsilon > 0$, there exists $\delta > 0$such that:

$$0 <| x - a |< \delta$$

implies:

$$| f(x) - L |< \epsilon$$

Q12. Why do limits matter for machine learning?

Because derivatives are defined using limits, and gradient-based optimization relies on derivatives/gradients.

Q13. Why does continuity matter for ML?

Continuity provides controlled local behavior. It is one of the mathematical regularity properties used in analyzing functions and optimization; differentiability then enables gradient-based methods.

Q14. What is the relationship between derivative and limit?

The derivative is defined as a limit:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Q15. What does gradient descent do?

It updates parameters in the direction opposite to the gradient:

$$w_{new} = w - \eta \nabla L(w)$$

to attempt to reduce the loss.

---

39. Common Interview Traps

Trap 1

"Not open means closed."

❌ False.

Trap 2

"Limit equals function value."

❌ Not necessarily.

That equality is required for continuity.

Trap 3

"Infimum must be inside the set."

❌ False.

Trap 4

"Compact means bounded."

❌ Incomplete.

For subsets of $\mathbb{R}$:

$$\boxed{\text{compact} = \text{closed} + \text{bounded}}$$

Trap 5

"If a function is continuous, it must be differentiable."

❌ False.

For example:

$$f(x) = |x|$$

is continuous at 0 but not differentiable there.

---

40. Implementation Cheat Sheet

Concept	**Practical implementation**
Limit	Evaluate $f(a+h)$for decreasing h
Derivative	$\big(f(x+h)-f(x)\big)/h$
Continuity	Compare numerical limit with $f(a)$
Optimization	Iteratively update parameters
Gradient descent	$w = w - \eta \nabla L(w)$
Bounds	Track min/max candidates
Infimum	Approximate greatest lower bound
Supremum	Approximate least upper bound

---

41. One-Minute Interview Revision

If you only have **one minute**, remember this:

OPEN SET

Every point has room around it.

CLOSED SET

Contains limits of its convergent sequences.

BOUNDED

Fits inside some finite interval.

MINIMUM

Smallest value actually inside set.

INFIMUM

Greatest lower bound; may not be inside.

MAXIMUM

Largest value actually inside set.

SUPREMUM

Smallest upper bound; may not be inside.

COMPACT

Closed + bounded in R.

LIMIT

Output approaches L as input approaches a.

CONTINUITY

Limit = actual function value.

DERIVATIVE

Limit of difference quotient.

GRADIENT

Multivariable generalization of derivative.

GRADIENT DESCENT

Move opposite the gradient to reduce loss.

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42. Final Mental Model for Your ML Interviews

The most valuable thing to understand is **not 20 isolated definitions**.

Understand this chain:

Sets → Neighborhoods → Sequences → Limits → Continuity → Derivatives → Gradients → Optimization →

The uploaded chapter itself ends with essentially this motivation: machine learning uses parameterized functions and seeks parameters that best fit observations; continuity and differentiability are important properties for making those functions workable for optimization.

Your 80/20 priority

P0 — Must know

1. Open vs closed sets
2. Neighborhood
3. Boundedness
4. Minimum/maximum
5. Infimum/supremum
6. Compactness
7. Limit intuition
8. Epsilon-delta definition
9. Limit vs function value
10. Continuity
11. Continuity vs differentiability
12. Derivative as a limit
13. Gradient descent connection

P1 — Know after P0

- sequence characterization of closed sets
- Bolzano-Weierstrass
- operations on limits
- operations preserving continuity
- Big-O / little-o
- special limits

- compactness and optimization

P2 — Don't spend excessive interview-prep time on initially

- lengthy formal proofs
- Cantor's axiom proof
- detailed topology formalism
- cardinality discussion
- elaborate theorem proofs

The chapter itself notes that some of these foundations are stepping stones toward the optimization concepts needed for ML rather than ends in themselves.